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14.5 + 2
16.5 = 161705

IT694

COMPUTER NETWORKS

60 mins IN-SEM EXAM I WINTER 2026 25 Marks

Instructions:

- Write your Student ID
- Answer all questions within the space provided
- Show all calculations clearly in the rough space provided.
- You will not need any additional sheet
- No questions will be entertained.

1. Host A is sending an enormous file to Host B over a TCP connection. Assume no packet loss and the timers never expire. Let the transmission rate of the link connecting Host A and the Internet be R bps. Suppose that the process in Host A is capable of sending data into its TCP socket at a rate S bps, where $S = 10 \times R$. Further suppose that the TCP receive buffer is large enough to hold the entire file, and the send buffer can hold only 1% of the file.

a) Will the receiver have any overflow issues? Justify [2 marks]

(2) No, the receiver will not have any buffer issue as already the transmission rate is less than sending rate as well as the TCP receive buffer is large enough to hold the entire file. So in worst case even if whole file is received the TCP buffer will hold it and send in segments to receiver.

b) What prevents the process in Host A from continuously passing data to its TCP socket at rate S bps? [2 marks]

(1/2) The send buffer prevents the Host A from continuously passing data as it can only hold upto 1% of the file. So when Host A tries to send it won't accept as segments are already in transmission as well as the buffer of TCP is also full rejecting to accept more data segments.

2. Use the table below to show the sequence of socket library calls made for a TCP connection by both client and server to set up their sockets and communicate. Assume the client sends a single segment of data to the server, the server receives it and responds with a single segment of data. You do not need to specify any arguments or return values (write `socket()` is sufficient). Show when each call issues and that the calls appear in the proper order. [10 marks]

Time	Client	Server
1	<code>socket()</code>	
2	<code>bind()</code>	
3	<code>connect()</code>	<code>socket()</code>
4	<code>accept()</code>	<code>connect()</code>
5	<code>listen()</code>	<code>send()</code>
6	<code>send()</code>	<code>receive()</code>
7	<code>close()</code>	<code>close()</code>
8		
9		
10		

(6+2) order is not proper

3. Suppose we send into the Internet two IP datagrams, each carrying a different UDP segment. The first datagram has source IP address A1, destination IP address B, source port P1, and destination port T. The second datagram has source IP address A2, destination IP address B, source port P2, and destination port T. Suppose that A1 is different from A2 and P1 is different from P2.

Assuming that both datagrams reach their final destination, will the two UDP datagrams be received by the same socket? Why or why not? [1 mark]

- Yes, both the UDP datagrams will be received by the same socket because they are being sent from different IP address (A1, A2) and from different ports (P1, P2) but their destination is same (IP: B, Dest: T)

4. What is the purpose of the IF-MODIFIED-SINCE field in an HTTP GET message? [1 mark]

- If a client has requested any file, the web cache server verifies with original web server whether the file has been updated or not. So, IF-MODIFIED-SINCE field in HTTP GET helps the cache server to know whether the file is updated or needs to be updated.

5. Suppose a TCP SYN message is sent from a client with IP address 128.119.40.186 and client port number 5345 to a server with IP address 41.123.7.94 and server port number 80 (HTTP)

- a) Once the TCP connection has been established, what will be the client-side IP address, client-side port number, server-side IP address and server-side port number of the TCP segment carrying the HTTP GET message. [2marks]

Client-side IP address : 128.119.40.186 (destination IP)
 Client-side port number : 5345 (destination port)
 Server-side IP address : 41.123.7.94 (source IP)
 Server-side port number : 80 (source port)

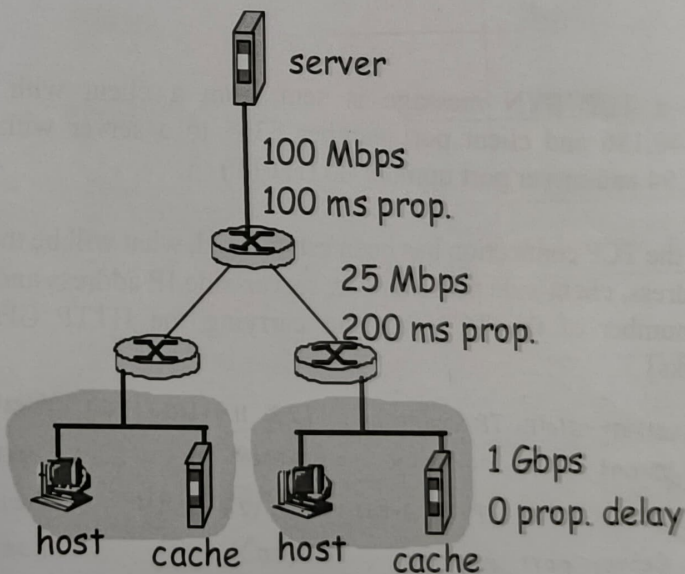
- b) Will the TCP SYN message and the HTTP GET message be directed to the same socket at the server? [2marks]

Yes, the TCP SYN message and HTTP GET message will be directed to the same socket at the server.

As TCP is just a connection protocol which facilitates the HTTP for request/response with the client.

So, the TCP ~~syn~~ carrying HTTP GET will go to port 80 (HTTP)

6. Consider the scenario in the figure below in which a server is connected to a router by a 100 Mbps link, with a 100 ms propagation delay. That router in turn is connected to two routers, each over a 25 Mbps link with a 200 ms propagation delay. A 1 Gbps link connects a host and a cache (when present) to each of these routers; this link, being a local area network, has a propagation delay that is essentially zero. All packets in the network are 10,000 bits long.



- a) What is the end-to-end delay from when a packet is transmitted by the server to when it is received at a host? Assume that there are no caches, that there is no queuing delay at a link, and that the node (router) packet-processing delays are also zero. [2marks]

→ $0.00125 + 0.01 + 0 = 0.01125$ - end-to-end delay from server to host

0

- b) Now assume that client hosts send requests for files directly to the server (i.e., the caches are off). What is the maximum rate at which the server can deliver data to a single client, assuming no other clients are making requests. [1 mark]

→ ^(0.035) 25mbps is the maximum rate at which the server can deliver data to single client why?

- c) Now suppose the caches are HTTP caches and are turned on. One client's HTTP GET is always first directed to its local cache. 50% of the requests can be satisfied by the local cache. What is the maximum rate at which this client can receive data in this scenario? [2marks]

→ If the client's request is a cache hit
maximum rate is 1Gbps

→ If the client's request is a cache miss
maximum rate is 25mbps